Esercizio.

Consider $\pi : RO(2^{<\omega}) \to RO(2^\omega), A \mapsto \bigcup_{q \in A} N_q$. Then π è isomorphism.

Dimostrazione.

Consider $k : 2^{<\omega} \to RO(2^\omega), s \mapsto N_s$ and $j : 2^{<\omega} \to RO(2^{<\omega}), s \mapsto Reg(\downarrow s)$. If we prove that k is completion map (i.e. order and incompatibility preserving with dense image) then we'll have that $k = \pi \circ j$ and π isomorphism.

- dense image: this is because $\{N_s : s \in 2^{<\omega}\}$ is a basis for 2^ω.
- order preserving: $s \leq t \Rightarrow s \supseteq t \Rightarrow N_s \subseteq N_t \Rightarrow k(s) \leq k(t)$.
- incompatibility preserving: $s \perp t \iff N_s \cap N_t = \emptyset$ is clear.

$\square$